

**Student ID (NOT your name):**

**Final Examination: QUESTION BOOKLET**

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- Do *NOT* open this question booklet until you are told to do so.
- Write your Student ID number (**NOT** your name) at the top of this page.
- Write your solutions in this booklet.
- No electronic devices are allowed during the exam.
- Be neat! If we can't read it, we can't grade it.
- You can treat any results from lecture or homework as "known," and use them in your work without rederiving them, but do make clear what result you're using. You do not need to explicitly check regularity conditions for the theorems from class that required them.
- For a multi-part problem, you may treat the results of previous parts as given (if you don't prove the result for part (a), you can still use it to solve part (b)).
- I have starred some parts which I believe are the most difficult, and which I expect most students won't necessarily be able to solve in the time allotted. They are generally not worth more points than the less difficult parts, so don't waste too much time on them until you're happy with your answers to the latter.
- Be careful to justify your reasoning and answers. We are primarily interested in your understanding of concepts, so show us what you know.

**Good luck!**

**1. The absent minded philosopher (20 points, 5 points / part).**

Your friend in the Philosophy department pays tens of thousands of dollars to run a Monte Carlo simulation on a large language model (LLM), getting independent real-valued observations  $X_1, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} P$  for  $N$  very large. He estimates  $\mu(P) = \mathbb{E}_P[X_i]$  via

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i,$$

getting a meaningfully positive value of  $\bar{X}$ . This is an exciting result because there is a consensus among philosophers of mind that  $\mu(P)$  precisely quantifies the degree to which the LLM experiences true consciousness.

Just one problem: he only recorded  $\bar{X}$  and deleted the data. Now a pesky referee is asking him for error bars!

Let  $\sigma^2(P) = \text{Var}_P(X_i)$  denote the variance of a single observation. Assume throughout the problem that  $\sigma^2(P) < \infty$ . Let  $\tau^2 = \frac{\sigma^2}{N}$  denote the variance of  $\bar{X}$ .

You tell him don't worry, he can just re-run his estimator on a new, smaller sample  $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} P$ , to get error bars.

We'll consider an asymptotic regime where  $N \rightarrow \infty$  and the small sample size  $n(N) \rightarrow \infty$  as well, but possibly not at the same rate (i.e., it could be that  $n(N)/N \rightarrow 0$ ). If you apply asymptotic theorems from class, be sure to say which ones you use at each step.

- (a) Give a finite-sample unbiased estimator  $\hat{\tau}^2(Y)$ , based on  $Y$  only, for  $\tau^2$ . Show that it is unbiased, and that it is consistent in the sense that  $\hat{\tau}^2/\tau^2 \xrightarrow{P} 1$ .
- (b) The referee is asking for a confidence interval too. Define

$$C(\bar{X}, Y) = [\bar{X} - \hat{\tau} z_{\alpha/2}, \bar{X} + \hat{\tau} z_{\alpha/2}],$$

where  $\hat{\tau} = \sqrt{\hat{\tau}^2}$  and  $z_{\alpha/2}$  is the upper  $\alpha/2$  quantile of a  $N(0, 1)$  random variable. Show that the coverage probability of  $C$  converges to  $1 - \alpha$  asymptotically.

- (c) Assume (for this part **only**) that  $P$  is  $N(\mu, \sigma^2)$ , for unknown  $\mu$  and  $\sigma^2$ . Show that if we replace the quantile  $z_{\alpha/2}$  with a quantile  $c_{\alpha/2}$  of some other distribution, we'll get an *exact* confidence interval in finite samples. What distribution should we use?
- (d) We can actually do a little better than  $\bar{X}$  now that we have the additional data. Give an unbiased estimator based on  $(\bar{X}, Y_1, \dots, Y_n)$  with smaller variance

than  $\bar{X}$ , and give its variance. For full credit, you should give the *best* such estimator (smallest variance) but you don't need to show it is the best.

**Problem 1 answers continued (1):**

**Problem 1 answers continued (2):**

## 2. Poisson with additive offsets (30 points, 5 points / part).

Some useful facts for this problem:

- The  $\text{Pois}(\lambda)$  distribution for  $\lambda > 0$  has probability mass function

$$\frac{\lambda^x e^{-\lambda}}{x!}, \quad \text{for } x = 0, 1, \dots$$

It has mean  $\lambda$  and variance  $\lambda$ .

- If  $X_i \stackrel{\text{ind.}}{\sim} \text{Pois}(\lambda_i)$  then  $\sum_i X_i \sim \text{Pois}(\sum_i \lambda_i)$ .
- For  $w = (w_1, \dots, w_n)$  and  $\lambda = (\lambda_1, \dots, \lambda_n)$ , with all  $w_i, \lambda_i > 0$ , let  $Q_{\lambda, w}$  denote the distribution of  $\sum_i w_i X_i$ , where  $X_i \stackrel{\text{ind.}}{\sim} \text{Pois}(\lambda_i)$ .

Assume we observe  $X_i \stackrel{\text{ind.}}{\sim} \text{Pois}(a_i + \theta)$  for  $i = 1, \dots, n$ , with **known offsets**  $a_1, \dots, a_n > 0$ , and unknown parameter  $\theta > -\min_i a_i$ .

In both parts below, where I ask you to give a test, it is good enough to (1) define a test statistic and (2) define the rejection threshold in terms of the upper  $\alpha$  quantile of a specific distribution. Don't worry about randomizing at the boundary.

- (a) Consider the estimator

$$\delta(X) = \frac{1}{n} \sum_i (X_i - a_i).$$

Show that it is unbiased and give its variance.

- (b) Show that  $\delta(X)$  is UMVU in the equal-offset case where  $a_1 = \dots = a_n = a$ , for some value  $a > 0$ .
- (c) Now assume that not all of the  $a_i$  values are equal. Find the maximum likelihood estimator  $\hat{\theta}_{\text{MLE}}$  for  $\theta$  as a solution to an equation  $f(\hat{\theta}; X_1, \dots, X_n, a_1, \dots, a_n, n) = 0$ . (You do not need an explicit form for  $\hat{\theta}$  but you should have an explicit form for  $f$ ).
- (d) Define  $\sigma^2$  as the asymptotic variance of  $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta)$ . Show that  $\sigma^2/n$  is no larger than the variance of  $\delta(X)$ , and strictly smaller in the unequal-offset case.

Don't worry about the fact that the  $X_i$  values are not i.i.d.; just assume the MLE has the asymptotic distribution prescribed by the usual theory.

**Hint:** Note the Cauchy-Schwarz inequality implies that

$$\left(\sum_i z_i\right)\left(\sum_i z_i^{-1}\right) \geq n^2 \quad \text{for } z_1, \dots, z_n > 0,$$

with equality if and only if all  $z_i$  are equal. You may use this result without proof.

- (e) Suppose that we want to test  $H_0 : \theta = 0$  vs  $H_1 : \theta = \theta_1$ , for a **specific value**  $\theta_1 > 0$ . What test should we use?
- (f) (\*) Suppose we want to test  $H_0 : \theta \leq 0$  vs  $H_1 : \theta > 0$ . Suggest a test that is valid in finite samples, and that is as powerful as possible against values of  $\theta$  that are barely larger than 0. For full credit, explain why your test is valid in finite samples.

**Problem 2 answers continued (1):**

**Problem 2 answers continued (2):**

**Problem 2 answers continued (3):**

**3. Inference on a discrete CDF (25 points, 5 points / part).**

Let  $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P$  be a random sample from an unknown distribution  $P$  supported on  $\{1, 2, \dots, 9\}$ . Define the cumulative distribution function

$$F(k) = \mathbb{P}(X_i \leq k), \quad k = 1, \dots, 9,$$

and the empirical CDF

$$\hat{F}_n(k) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq k\}.$$

Note that  $\hat{F}_n(9) = F(9) = 1$  deterministically. Throughout this problem, we therefore work with the vectors

$$\hat{\mathbf{F}} = (\hat{F}_n(1), \dots, \hat{F}_n(8)), \quad \mathbf{F} = (F(1), \dots, F(8)).$$

This problem asks you to come up with asymptotic tests and confidence regions. You do not need to prove they have the right asymptotic coverage, just give the right tests and regions in terms of data and quantiles of well-specified distributions.

(a) Show that

$$\sqrt{n}(\hat{\mathbf{F}} - \mathbf{F}) \Rightarrow N_8(0, \Sigma(\mathbf{F})),$$

and give an explicit expression for the covariance matrix  $\Sigma(\mathbf{F})$ . (Your answer should give  $\Sigma_{k\ell}$  as a function of  $F(k)$  and  $F(\ell)$ .)

(b) Using the asymptotic distribution from part (a), give an asymptotic level- $\alpha$  test of

$$H_0 : \mathbf{F} = \mathbf{F}_0 \quad \text{vs} \quad H_1 : \mathbf{F} \neq \mathbf{F}_0,$$

for a specific point null  $\mathbf{F}_0$ . You may assume without proof that  $\Sigma(\mathbf{F}_0)$  is invertible.

(c) By inverting the test from part (b), give an asymptotic  $(1 - \alpha)$  Wald confidence region for  $\mathbf{F}$ . Note the covariance matrix  $\Sigma(\mathbf{F})$  is unknown.

(d) Using the marginal asymptotic distribution of each  $\hat{F}_n(k)$ , construct a simultaneous  $(1 - \alpha)$  confidence band for  $F(1), \dots, F(8)$ . (A conservative band is acceptable.)

- (e) (\*) The random variables  $\hat{F}_n(1), \dots, \hat{F}_n(8)$  are positively correlated. Explain how the joint asymptotic normal distribution from part (a) can be used to obtain a strictly narrower simultaneous confidence band than the one in part (d). You do not need to give critical values explicitly, but you should specify clearly how they would be defined or approximated.

**Problem 3 answers continued (1):**

**Problem 3 answers continued (2):**

**4. Ising model (25 points, 5 points / part).**

Consider the one-parameter family of distributions on  $\{0, 1\}^d$  given by

$$p_\beta(x) = \frac{1}{Z(\beta)} \prod_{i < j} e^{\beta |x_i - x_j|},$$

where the normalizing constant is

$$Z(\beta) = \sum_{x \in \{0, 1\}^d} \prod_{i < j} e^{\beta |x_i - x_j|}.$$

This is called the *Ising model*. Let  $S(x) = \sum_{i=1}^d x_i$  denote the sum of  $x \in \{0, 1\}^d$ .

For  $x \in \{0, 1\}^d$ , let  $S(x) = \sum_{i=1}^d x_i$  and  $T(x) = S(x)(d - S(x))$ .

- (a) Show that  $\{p_\beta : \beta \in \mathbb{R}\}$  is a one-parameter exponential family and show that  $T(X)$  is a complete sufficient statistic.
- (b) Is  $S$  a sufficient statistic? Is it a complete sufficient statistic? Justify your answers.
- (c) Let  $A(\beta) = \log Z(\beta)$  and consider the prior family on  $\beta$  given by

$$\lambda_{\eta, \kappa}(\beta) \propto \exp\{\eta \beta - \kappa A(\beta)\}, \quad \kappa > 0.$$

Show that this family is conjugate for the model, and give the parameters  $(\eta_{\text{post}}, \kappa_{\text{post}})$  for the posterior after observing a single draw  $X \sim p_\beta$ .

- (d) Give a level- $\alpha$  UMP test of

$$H_0 : \beta \leq 0 \quad \text{vs} \quad H_1 : \beta > 0.$$

It is enough to (i) specify a test statistic, and (ii) define the rejection threshold in terms of a quantile of a precisely specified distribution. (You do not need to randomize at the boundary.)

- (e) (\*) Now introduce a nuisance parameter  $\alpha \in \mathbb{R}$  and consider the two-parameter family

$$p_{\beta, \alpha}(x) \propto \exp\{\beta T(x) + \alpha S(x)\}, \quad x \in \{0, 1\}^d.$$

Suppose we observe i.i.d. samples  $X^{(1)}, \dots, X^{(n)} \sim p_{\beta, \alpha}$ . Give an *exact* level- $\alpha$  test of

$$H_0 : \beta \leq 0 \quad \text{vs} \quad H_1 : \beta > 0$$

that remains valid for all values of the nuisance parameter  $\alpha$ . (Again it is enough to specify a test statistic and a rejection threshold via a quantile of a precisely specified distribution.)

**Problem 4 answers continued (1):**

**Problem 4 answers continued (2):**

**Problem 4 answers continued (3):**